INCH-POUND

MIL-T-50752B (AR)
04 January 1993
SUPERSEDING
MIL-T-50752A (AR)
30 July 1984

MILITARY SPECIFICATION

TEST SET, SUBASSEMBLY MX-8046A/TPM-22A (RECEIVER/TRANSMITTER TESTER)

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Test Set, Subassembly MX-8046A/TPM-22A, hereinafter referred to as receiver/transmitter tester, a part of the Radar Test Set-AN/TPM-22A. The receiver/transmitter tester makes receiver measurements and provides power and control voltages to test some of the subassemblies in unit 2 of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in-the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of
MXL-G-50758 - Generator, Pulse Assembly: 10556650

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

STANDARDS
AMSC N/A

FSC 4931

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.4)

u. s. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548158 - 60 MHz Source
10548160 - Mixer Preamplifier X-Band
10548182 - X-Band Frequency Source
10556651 - Power Source
10558406 - IF Amplifier
11739273 - Test Set, Subassembly MX-8046A/TPM-22A
11739281 – Schematic
11739284 - Test Cable

SPECIAL PACKAGING INSTRUCTION

PS 11739273 - Packaging of Test Set Subassembly MX-8046A/TPM-22A

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11739273, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the receiver/transmitter tester shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The receiver/transmitter tester shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per **axis.**
The sweep rate shall be 3.75 minutes ± 10% per decade
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The receiver/transmitter tester shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The receiver/transmitter tester shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg) .

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

3.3.3.2 Non-operating. The receiver/transmitter tester shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The receiver/transmitter tester shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg F and +155 deg F.

3.3.4.2 Non-operating (storage). The receiver/transmitter tester shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg F for a period of 24 hours and +160 deg F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the receiver/transmitter tester.

3.5 Performance.

3.5.1 Power Switch S1.

3.5.1.1 The voltages in Table I (Receiver/transmitter voltage inputs) are required by the receiver/transmitter.

TABLE I. Receiver/transmitter voltage inputs.

J13	Voltage
A	+28 Vdc ± 0.1%
P	+28 Vdc ± 0.1%
B	+28 Vdc Ret
N	+28 Vdc Ret
T	-33 Vdc ± 0.5%
U	-33 Vdc Ret
E	+15 Vdc ± 0.5%
G	-15 Vdc ± 0.5%
F	±15 Vdc Ret
R	+24 Vdc ± 4 Vdc
V	+24 Vdc ± 4 Vdc control in
W	+24 Vdc control out
S	+24 Vdc Ret
X	Frame Ground

Source: https://assist.dla.mil Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

3.5.1.2 OFF. When S1 is in normally closed position the POWER and OFF legend shall be illuminated. All other indicators shall be extinguished.

3.5.1.3 ON. When S1 is in normally open position and POWER and ON legend of S1 shall be illuminated and the following conditions exist:

PULSE GENERATOR and NOISE FIGURE legends are extinguished when their corresponding switches are in the normally closed position.

b. OFF legend on RADIATE switch shall be illuminated in its normally closed position.

c. INTERNAL legend on INTERNAL/EXTERNAL switch shall be illuminated in its normally closed position.

d. IF AMPLIFIER on/off switch indicator shall be extinguished in its normally closed position.

e. See Table 11 (Voltage requirements) for the following voltage requirements:

TABLE II. Voltage requirements.

FROM (+)	TO (-)	VOLTAGE
J5-1	J5-5	+28 V ± 0.1%
J7-2	J7-4	+28 V ± 0.1%
J7-7	J7-6	+33 V ± 0.5%
J7-1	J7-3	+15 V ± 0.5%
J7-3	J7-5	+15 V ± 0.5%
C2	FL 8	+24 V ± 4 Vdc
J4-U	J4-S	+33 V ± 0.5%
J4-N	J4-K	+15 V ± 0.5%
J4-K	J4-M	+15 V ± 0.5%
J4-X	J4-W	+28 V ± 0.1%

f. When there is no voltage on pin R of J-13 and all other voltages are applied in accordance with Table I, all legends except S-1 shall be extinguished.

FROM (+)	TO (-)	VOLTAGE
J4-V	J4-W	+28 V ± 0.1% Vdc
J4-G	J4-T	+24 V ± 4 Vdc
J4-G	J4-C	+24 V ± 4 Vdc
J4-G	J4-H	+24 V ± 4 Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

g. STALO legend and OFF legend on STALO switch shall be illuminated in its normally closed position.

3.5.2 RADIATE switch S2.

a. With conditions stated in 3.5.1.3 and S2 in normally open position, ON legend of S2 shall be illuminated and OFF be extinguished. The ON legend of S2 shall be illuminated for 2 minutes ± 12 seconds after which RADIATE and ON legend shall be illuminated and 24 Vdc ± 4 Vdc shall be present between J4-R(+) and J4-F(RET).

b. The RADIATE light should extinguish when the NOISE FIGURE switch is in the normally open position and the NOISE FIGURE legend is illuminated.

c. The time delay cycle shall be reactivated when the NOISE FIGURE switch is placed in the normally closed position and the NOISE FIGURE legend extinguished.

3.5.3 NOISE FIGURE switch S3. With conditions stated in 3.5.1.3 the legend of S3 shall not be illuminated when S3 Is in normally closed position. When S3 is in normally open position the entire NOISE FIGURE legend shall be illuminated.

3.5.4 STALO switch S4. With conditions stated in 3.5.1.3 the STALO and ON legend of S4 will be illuminated when S4 is in the normally open position.

3.5.5 PULSE GENERATOR/PULSE/SIG switch S5.

a. With conditions stated in 3.5.1.3 and the GEN DRIVER switch S10 in the normally open (off) position, the PULSE GEN and SIG legend of S5 will be illuminated when S5 is in the normally open position.

b. When the GEN DRIVER switch S10 is in the normally closed position the entire legend of S5 will be illuminated.

3.5.6 INTERNAL/EXTERNAL switch S6.

a. With conditions stated in 3.5.1.3 the INTERNAL legend of S6 will be illuminated when S6 is in the normally closed position.

b. The EXTERNAL legend of S6 will be illuminated and the INTERNAL legend will be extinguished when S6 is in the normally open position.

3.5.7 IF AMPLIFIER PWR switch S11.

a. With conditions stated in 3.5.1.3 the S11 indicator will be extinguished in the normally open position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

b. The S11 indicator will be illuminated in the normally closed position.

3.5.8 IF AMPLIFIER A3. With conditions stated in 3.5.1.3 and S11 in the normally closed position, the gain of the amplifier shall be not less than 62 db min and not greater than 66 DB.

3.5.9 Attenuator AT1 and DC block.

3.5.9.1 Attenuator AT1 shall provide a continuously variable attenuation of the signal present at output jack J3. The amount of attenuation shall be controlled by attenuation control and indicated on control dial in db and meet the following requirements:

Insertion loss - at O db the dial setting shall be greater than .7 db.

b. Attenuation – O to 50 db ± 2%.

3.5.9.2 The dc output level at output jack J3 shall be Of0.001 Vdc.

3.5-10 AGC ADJ R1. Voltage between J19 (center contact) and ground shall be O ± 0.1 Vdc to 6 Vdc (minimum), variable by operation of AGC ADJ control R1 with conditions stated in 3.5.1.3.

3.5.11 PULSE GENERATOR A5. With conditions stated in 3.5.5.1, the PULSE GENERATOR shall provide the adjustible pulse present at output jack J6 and meet the following requirements with S9 in INTERNAL position (NC).

a. pulse width 0.2 to 0.4 usec at the 50% point.

b. output : Output shall be 6.5 ± 1.5 V at maximum setting of level ADJ control R3 (see Figure 1) . Peak output shall be .55 V maximum at minimum setting of R3 (see Figure 2).

c. Frequency: HIGH frequency cutoff should be a repetition rate of 26 KHz minimum low frequency cutoff should be a maximum PRF of 4 KHz.

When an external pulse is applied at EXTERNAL SYNC jack, HIGH frequency cutoff shall be a minimum rate of 99 KHz; LOW frequency cutoff should be a maximum rate of 1 KHZ.

3.5.12 Insertion loss between ATTENUATOR IN jack (Jl) and PULSE OUT jack (J3). With conditions stated in 3.5.6b, MOD MODE switch S8 in the ON position and a frequency between 9.14 GHz and 9.245 GHz inserted at Jl, the maximum insertion loss shall be 4.0 db maximum measured at J3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

3.5.13 MICROWAVE POWER SOURCE. With conditions stated in 3.5.4, the power source shall meet the following requirements:

a. Power output : 100 mW minimum, 300 mW maximum.

b. Channel output frequency:

Channel Selector	Frequency (MHz)	Prequency Tolerance
1	9145.000	9144.543 to 9145.457 MHz
2	9153.000	9152.542 to 9153.458 MHz
3	9161.000	9160.542 to 9161.458 MHz
4	9169.000	9168.541 to 9169.458 MHz
5	9177.000	9176.541 to 9177.459 MHz
6	9185.000	9184.541 to 9185.459 MHz
7	9204.936	9204.475 to 9205.396 MHz
8	9212.936	9212.475 to 9213.386 MHz
9	9220.936	9220.475 to 9221.397 MHz
10	9228.936	9228.474 to 9229.397 MHz
11	9236.936	9236.474 to 9237.297 MHz
12	9244.936	9244.473 to 9245.398 Mhz

Source: https://assist.dla.mil - Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

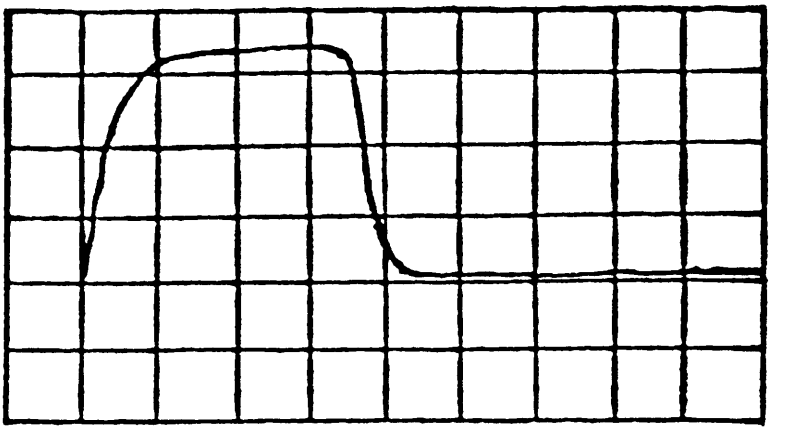


FIGURE 1. Voltage output.

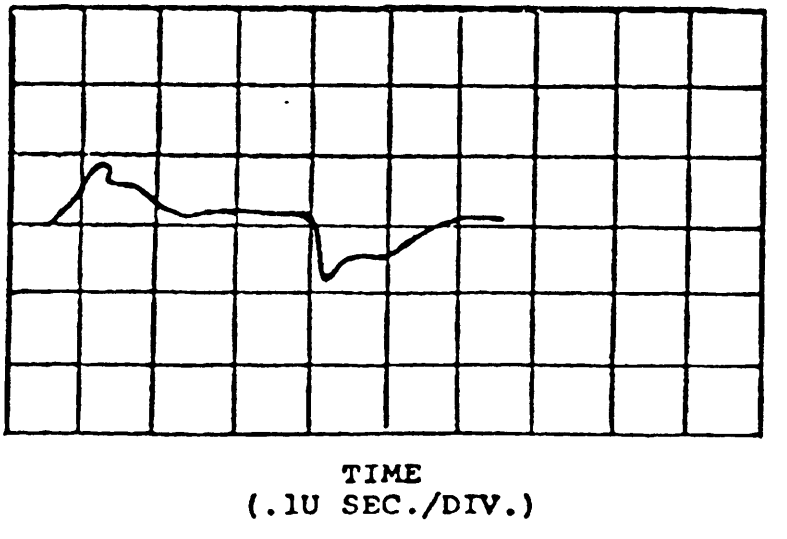


FIGURE 2. Voltage output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

From 1 KHz to 110 KHz there shall be no spurious noise greater than -50 db below the carrier.

d. With a square wave frequency between 600 Hz and 20 KHz and 4 V peak-to-peak applied to MOD in jack (J23), spurious FM noise shall be no greater than -so db below the carrier.

3.5.14 Microwave chassis (A2) insertion loss and attenuation. With the power measurement in 3.5.13, the microwave chassis (A2) shall meet the following requirements:

a. With conditions stated In 3.5.4, the power output at STALO MONITOR jack J9 shall be 20 ± 1 db down from the specification in 3.5.13a.

b. with conditions stated in 3.5.4 and MODE switch S8 in ON position, the power output at spectrum analyzer jack (J24) shall be 32 ± 1 db down from the power measurement in 3.5.13a.

c. With conditions stated in 3.5.4 and MODE switch S8 in ON position, the power output at PULSE OUT jack J3 shall be no greater than 4 db down from the power measurement in 3.5.13a.

3.5.15 SPST Modulator (A2K2).

With conditions stated in 3.5.4 and MODE switch S8 in OFF position, the power output at PULSE OUT jack J3 shall be no greater than -4o db down from the measurement in 3.4.13a.

b. With conditions stated in 3.5.4, 3.5.5b and 3.5.6a, MOD MODE switch to PULSE, PULSE SYNC switch to INT and PRFs of 25 KHz, 20 KHz and 10 KHz (R2 Adjust), the following frequency spectra will be obtained for all 12 channels (see Figures 3a, b and c).

3.5.16 NOISE SOURCE reference output (A2A1). With conditions stated in 3.5.3 the noise source output at PULSE OUT jack J3 shall be 25.0 ± 0.5 db above KTB after miniature microwave attenuator (A2ATl) on microwave chassis has been adjusted and attenuator set to 0 db on front panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

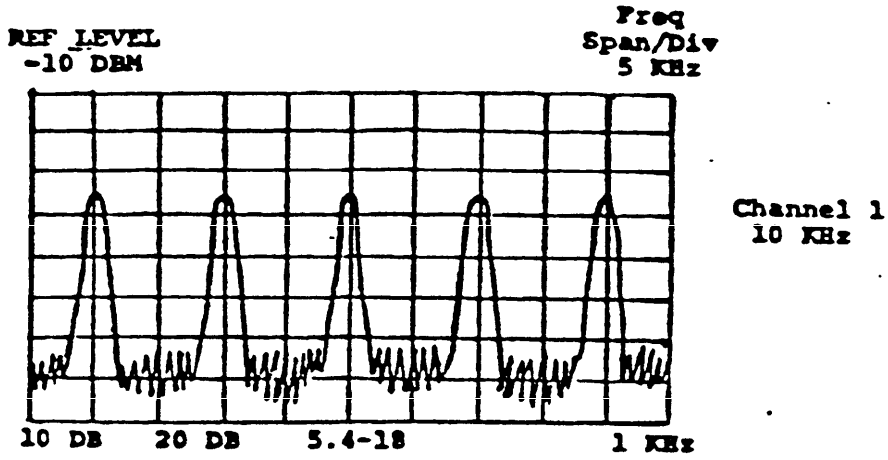


FIGURE 3a. Frequency spectrum.

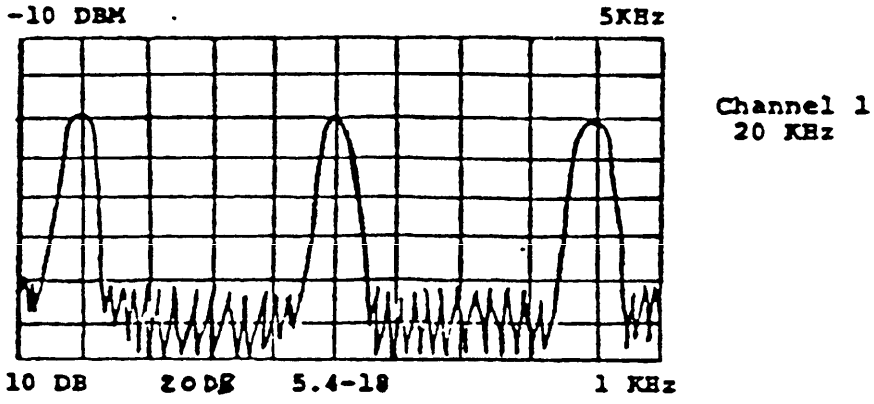


FIGURE 3b. Frequency spectrum.

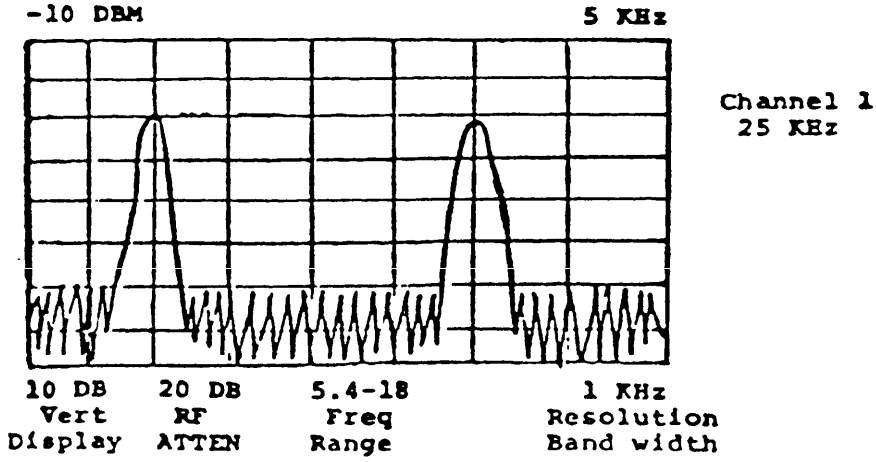


FIGURE 3c. Frequency spectrum.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:36Z
Check the source to verify that this is the current version before use.

3.5.17 MOD trigger Jll, J15, J5-A1. With a load of 95 ohms ± 5% at J15 and a square wave input signal of 5 volt p-p maximum with a PRF of 20 KHZ minimum, there will be no phase distortion between the input pulse at Jll and the output pulse at J15 and J5-A1.

3.5.18 60 MHz CW J12 and J16. With a load of 50 ohms ± 5% at J16 and an input signal of 60 MHz CW, there will be no amplitude or phase distortion between J12 (input) and J16 (output).

3.5.19 TO trigger J17, J14, J18. With a load of 50 ohms at J17 and J18 and 10 volts peak-to-peak at J14 with PRF at 20 KHz minimum and a pulse width of 0.5 usec, there will be no phase or amplitude distortion between the input pulse at J14 and the output pulse at J17 and J18.

3.5.20 +24 Vdc control out. With the voltages in Table I applied, there will be +24 ± 4 Vdc at J13-W. (S1 is in normally open, 3.5.1.3.)

3.6 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance-with the technical provisions herein (see 2.3).

3*7 soldering. All soldering shall be in accordance with the requirements of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection **requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.**

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Receiver/Transmitter Tester	11739273	1

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3*3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:36Z
Check the source to verify that this is the current version before use.

product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the-type of inspection procedure which-is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment such as statistical process control, tool control, other types of sampling procedures, etc. , may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. when required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:36Z
Check the source to verify that this is the current version before use.

4.4.2.3 Functional tests. The tests in Table III shall be performed on a 100% basis.

TABLE III. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One receiver/transmitter tester out of the first so produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table III and shall then meet the tests in Table IV. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE IV. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.6 Inspection equipment .

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error In the test equipment design must be considered as part of the prescribed product tolerance limit. Thu S , concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table V (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE V. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Voltage & resistance measuring device	0-40 Vdc ± .1%. Continuity measuring capabilities.
2.	Oscilloscope	Tektronix Type 454 or similar.
3.	60 MHz source	10548158.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE V. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
4.	Attenuator	80 db variable coaxial Jerrold AV-50-2 3 or similar.
5.	BNC TEE	3, with 2 jacks and one plug each.
6.	Pulse generator	PRF - 0 to 100 KHz. Voltage - 0 to +10 v. Pulse width 0.2 usec max to 5 usec min. 50 ohm impedance.
7.	Frequency counter	Hewlett Packard 5345A with plug in 5255A frequency converter or similar.
8.	Waveguide	x-band from 6 to 12 fiches.
9.	Spectrum analyzer	To 10 GHz 60 db range Tektronix 492 or similar.
10.	Signal generator	600 Hz to 25 KHz square wave and sine wave 5V p-p.
11.	Attenuator	x-band , 20 db fixed with N connectors.
12.	Power meter	Hewlett Packard Type 432A or similar.
13.	Noise tube	Standard, calibrated for 25 db noise above KTBn.
14.	Termination	BNC 95 Ohms.
15.	Termination	2 BNC 50 ohms.
16.	Probe	Standard oscilloscope probe, Tektronix or similar.
17.	X-band frequency source	Between 9.145 GHz and 9.245 GHz 10548182 with plug in crystal or power source 10556651.
18.	Mixer preamplifier x-band	10548160
19.	Power supply (de)	20-30 volts at 10 amperes, regulated to 0.1%.
20.	Stop watch	0-60 min $\pm$ 0.5 sec.
21.	Power supply (de)	25-40 volt at 5 amperes regulated to 0.5%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

TABLE V. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
22*	Power supply (de)	10-20 volt at 10 amperes regulated to 0.5%.
23.	Power supply (db)	20-30 volt at 10 amperes, regulated to 0.5%.
24.	Power supply (de)	20-30 volt at 10 amperes, regulated to 0.5%0
25.	Switch	SPST , contact rate 20 amp.
26.	Switch	SPST, contact rate 20 amp.
27.	Resistor	500 ohm, 10%, 2 watts.
20.	Test cable	11739284-1.
29.	IF amplifier	10558406 (part of MX-8046A) or equivalent.
30.	Attenuator, waveguide	Waveline 622, or similar.
31.	Adapter	Waveguide to N jack.
32.	Termination	50 ohm, 0.5 W, DC-10 GHz, with N connector.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph 11-2.1 apply. The receiver/transmitter tester shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the receiver/transmitter tester shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.7.1.2 Shock . This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the receiver/transmitter tester shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7*1.4 Temperature.

4.7.1.4.1 High (operating and non-operating. This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) . This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified In 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

4.7.3 Performance.

PANEL - Before proceeding with the performance tests, ascertain that all panel switches are in their de-energized positions, using an ohmmeter (item 1, Table V).

4.7.3.1 Power switch S1. Using a voltmeter (item 1, Table V) note that the following voltages exist between the following points. When power is supplied to tester (see Figure 4), switches (items 25 and 26, Table V) shall be closed.

FROM (+)	TO (-)	VOLTAGE
FL 3	FL 2	33 V ± 0.5%
FL 4	C 3	15 V ± 0.5%
C 3	FL 5	15 V ± 0.5%
FL 12	FL 8	24 V ± 4 Vdc
FL 7	FL 8	24 V ± 4 Vdc

Depress POWER switch S1 and note that the following takes place:

a. ON legend of POWER switch illuminates and OFF legend extinguishes.

b. INTERNAL legend of INTERNAL/EXTERNAL switch illuminates.

c. OFF legend of RADIATE switch illuminates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

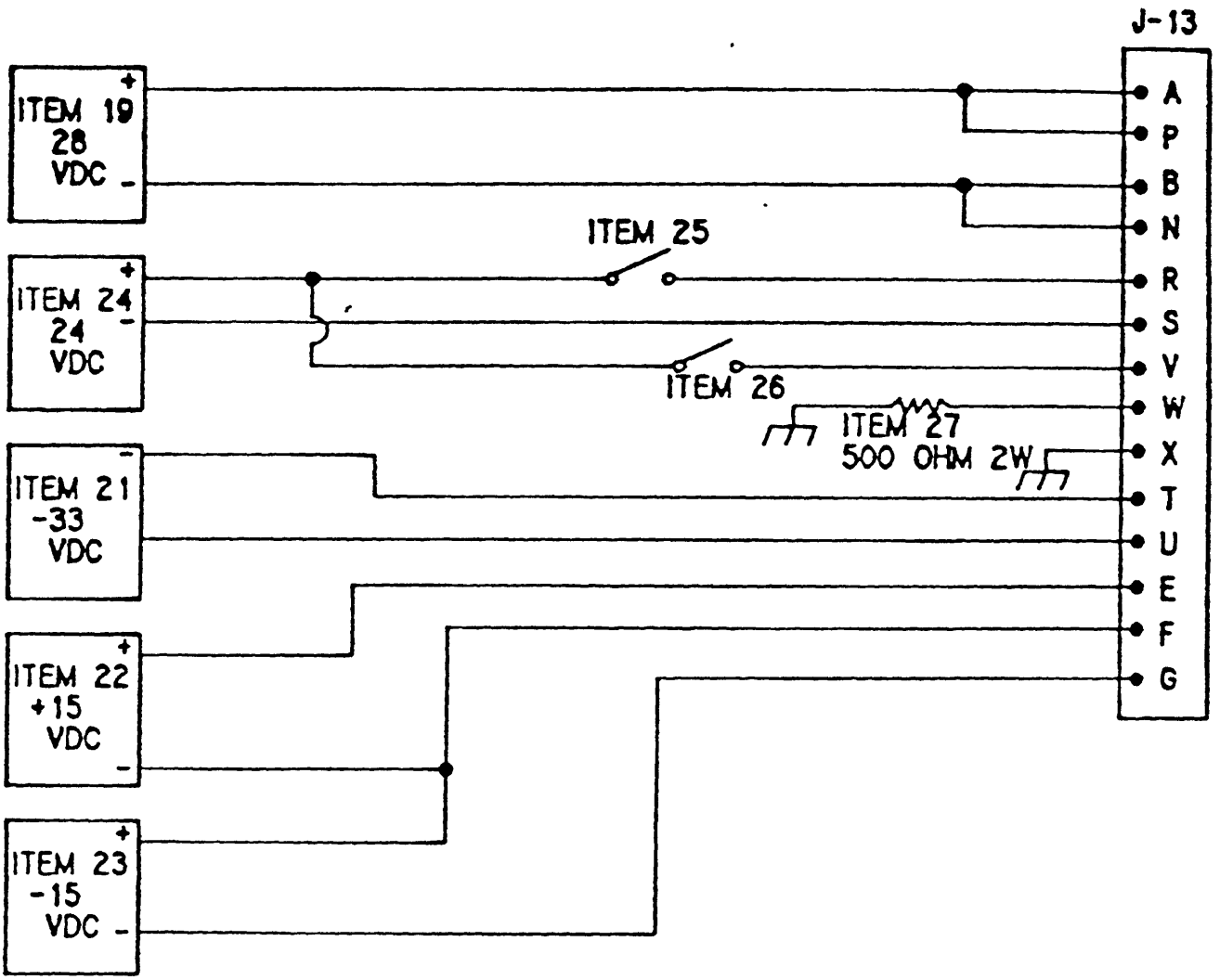


FIGURE 4. Dc power supplies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

d. STALO legend and OFF legend switch shall be illuminated. All other indicators remain extinguished.

e. 24 V ± 4 Vdc shall be present from C2(+) to FL 8(-).

With the voltmeter (item 1, Table V) measure the voltages listed in 3.5.1.3 at the points indicated. When switch (item 25, Table V) is open, the light will meet the requirements in paragraph 3.5.1.3f.

4.7.3.2 RADIATE switch S2. Depress the RADIATE switch and note that the ON legend illuminates and OFF legend extinguishes. After 2 minutes ± 12 seconds the RADIATE legend illuminates and a potential of 24 Vdc ± 4 Vdc appears between J4-R(+) and J4-F(-). Verify using stop watch (item 20, Table V)-

4.7.3.3 NOISE FIGURE switch S3. With RADIATE and ON legends of S2 illuminated depress NOISE FIGURE switch S3. Note that the following takes place:

a. NOISE FIGURE legend of S3 illuminates.

b. RADIATE legend of S2 extinguishes. Depress S3 to extinguish legend. After 2 minutes ± 12 seconds the RADIATE legend shall reilluminate.

4.7.3.4 STALO switch S4. Depress S4 (STALO switch) and note that ON legend illuminates. Depress again and note that legend extinguishes.

4.7.3.5 PULSE GENERATOR switch S5. Note that the GEN DRIVER switch S10 is in the OFF position. Depress S5 and note that the PULSE GEN and SIG legends illuminate. Set S10 to the ON position and note that DRIVE legend of S5 illuminates.

4.7.3.6 INTERNAL-EXTERNAL switch S6. Depress S6 and note that the INTERNAL legend extinguishes and EXTERNAL legend illuminates.

4.7.3.7 IF AMP switch S11. Depress S11 and note that the green indicator illuminates. At completion of panel check out, set all switches in the positions such that the conditions stated in 4.7.3.1 exist.

4.7.3.8 AMPLIFIER A3. Connect as shown in Figure 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

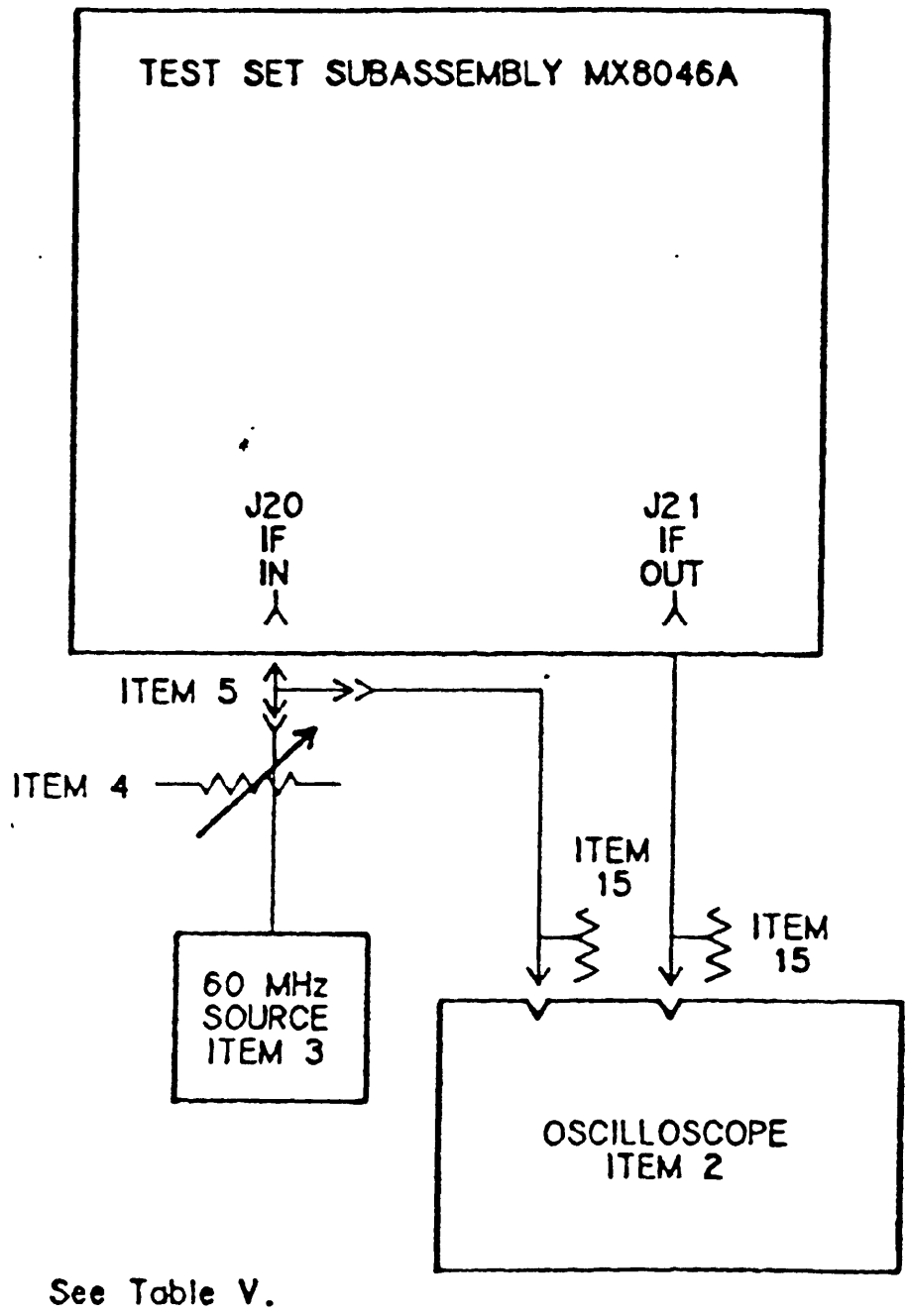


FIGURE 5. Test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

Oscilloscope settings:

Gain -CH l - 5 mV/div

CH 2 - 10 V/div

Channel select – Alt

Sync - int.

Sweep - 20 usec/div

Attenuate signal from 60 MHz oscillator with attenuator (item 4, Table V) until the signal is within dynamic range of IF AMP (i.e., no AGC action).

Depress IF AMP switch to illuminate green indicator. Channel 2 signal shall indicate a gain of 64 ± 2 db over signal on channel 1.

4.7.3.9 AGC adjust R1. Connect positive lead of voltmeter (item 1, Table V) to center contact of AGC jack of MX-8046A panel, and connect negative lead to frame. Voltage shall vary continuously from NMT 0.1 Vdc at minimum settings of AGC to NLT + 6 Vdc at maximum setting.

4.7.3.10 PULSE GENERATOR AS. Using a standard coaxial cable connect PULSE GEN jack to charnel 1 of oscilloscope (item 2, Table V). Terminate at oscilloscope with 50 ohms (Item 15, Table V).

Oscilloscope settings:

Gain - 2 V/div

Sweep - 50 usec/div

Sync - int.

Depress PULSE GEN switch S5 to illuminate PULSE GEN and SIGNAL legends, set GENERATOR DRIVER toggle switch to OFF position and note that legend of PULSE GEN DRIV is not illuminated. Set PULSE SYNC toggle switch to INT. Pulses visible shall be 5 to 8 volts peak-to-peak at maximum setting of LEVEL ADJ control and NMT 0.5 V peak-to-peak at minimum setting.

Adjust FREQ ADJ until the number of pulses visible on oscilloscope reaches a minimum. At this setting, the time between pulses shall be NLT 250 usec. Adjust FREQ ADJ until the number of pulses visible reaches a maximum. Set the scope sweep to 10 usec/div. At this setting the time between pulses shall be NMT 38 usec. Set oscilloscope sweep to 0.2 usec/div and locate a pulse. The pulse visible shall be 0.2 to 0.4 usec at the 50% point. Connect a BNC TEE (item 5, Table V) to output jack of pulse generator (item 6, Table V) and connect TEE jacks to SYNC jack of MX8046A and charnel & jack of oscilloscope. Set oscilloscope sweep to 50 usec/div and

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

channel selector to ALT. Set PULSE WIDTH on pulse generator (item 6, Table V) to approximately 0.3 usec. Set SYNC toggle switch on MX-8046A to EXT. Note that with a PRF of less than 5 pulses per division, the (100 KHz) pulses on channel 1 and channel 2 are synchronize ed.

4.7.3.11 MICROWAVE POWER SOURCE.

Power source frequency measurement. Connect J3 (TRANSMIT PULSE OUT) to frequency counter (item 7, Table V) using W80 (item 28, Table V). Depress STALO switch S4, set MOD MODE switch to ON and verify that the frequencies generated by the power source are the same as those listed in 3.5.13 for all twelve channels.

b. Power source spectrum analysis. Disconnect W80) (item 28, Table V) from frequency counter (item 7, Table V) and connect to spectrum analyzer (item 9, Table V). With STALLO switch depressed, ON legend illuminated, INTERNAL/EXTERNAL switch (S6) to INTERNAL and MOD MODE switch set to ON, CW signal shall appear as a narrow spike with no sideband noise greater than 50 db below the carrier (see Figure 6).

Connect signal generator (item 10, Table V) to O MOD -jack on tester panel. With conditions as described in b above and a square wave with a frequency of 600 Hz, 4 V peak-to-peak generated by frequency generator, there shall be side-band noise visible on spectrum analyzer screen (see Figure 7).

d. Repeat step c with 20 KHz (see Figure 8).

e. Using signal generator (item 10, Table V) insert a 5 KHz sine wave 10 V peak-to–peak into Phase 0 MOD jack on tester panel. A frequency modulated (FM) signal shall be visible on analyzer screen (see Figure 9).

f. Disconnect signal generator and set MOD MODE switch to PULSE l Depress PULSE-GEN/SIG/DRIVE switch to illuminate PULSE GEN and SIG legends. Set GEN DRIVER switch to ON and PULSE SYNC switch to INTERNAL. Spectrum visible on analyzer shall be that charac-teristic of a pulse modulated signal (see Figures 10a and 10b).

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:36Z
Check the source to verify that this is the current version before use.

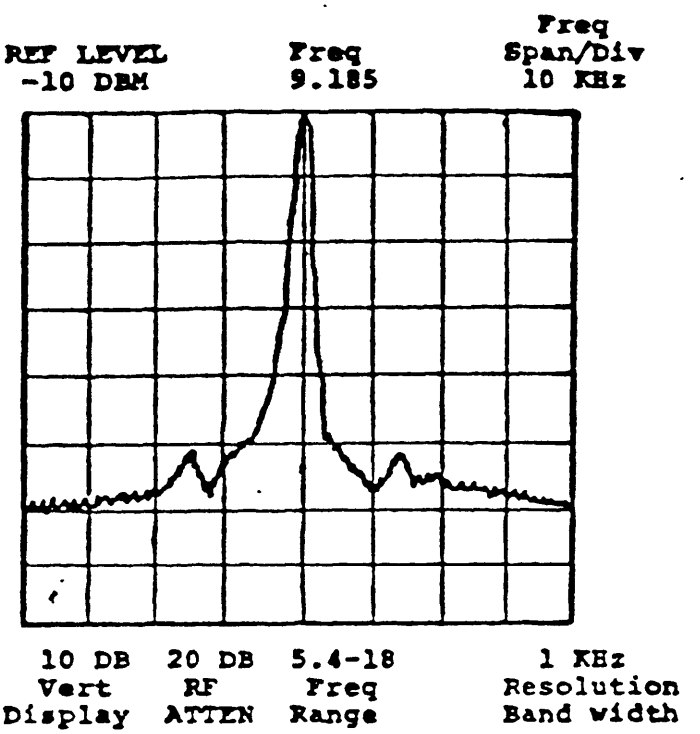


FIGURE 6. Frequency display.

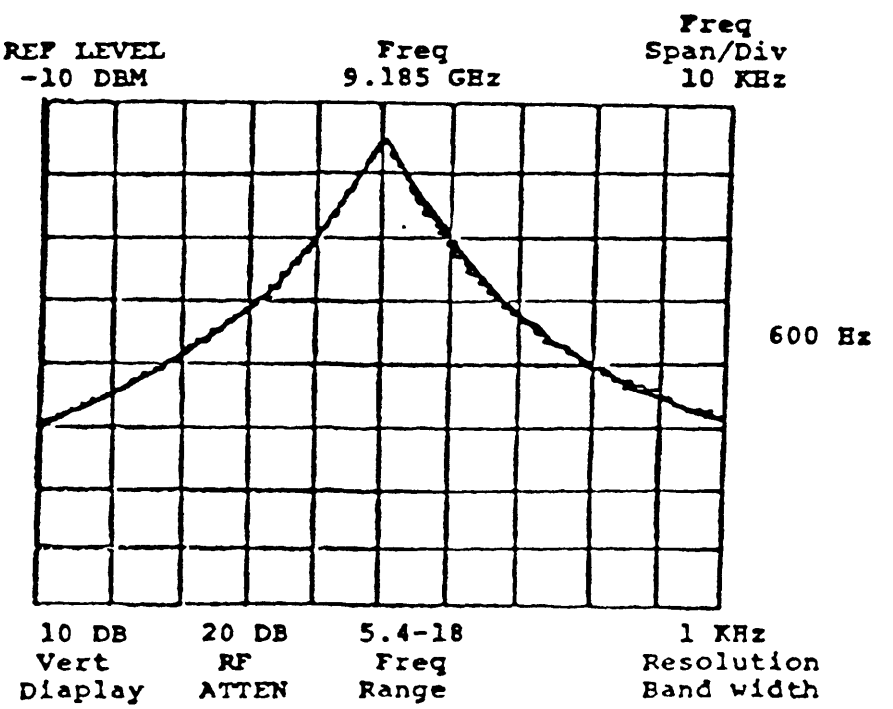


FIGURE 7. Frequency display with side-band noise.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:36Z
Check the source to verify that this is the currentversion before use.

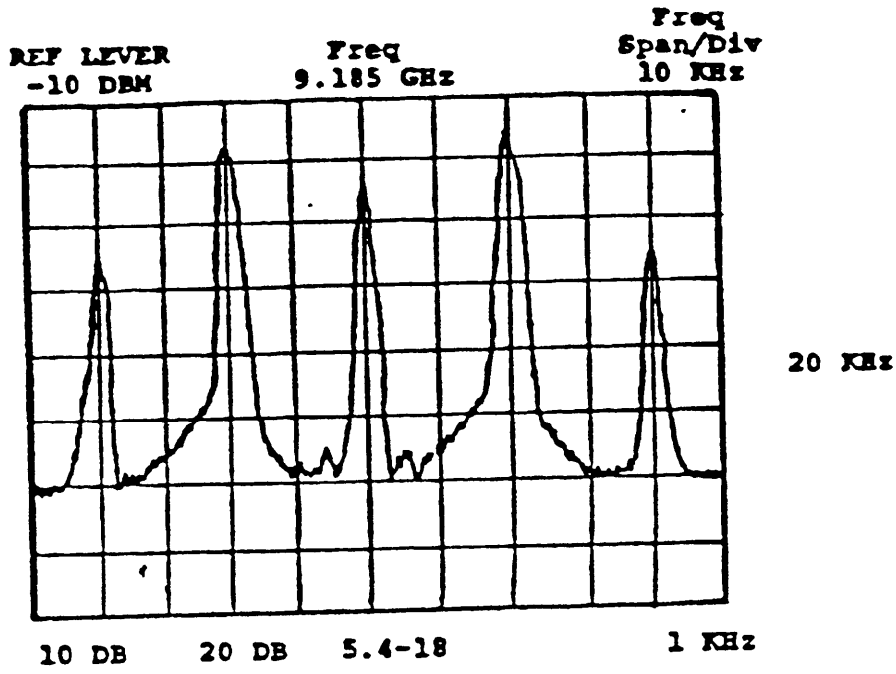


FIGURE 8. Frequency modulated signal.

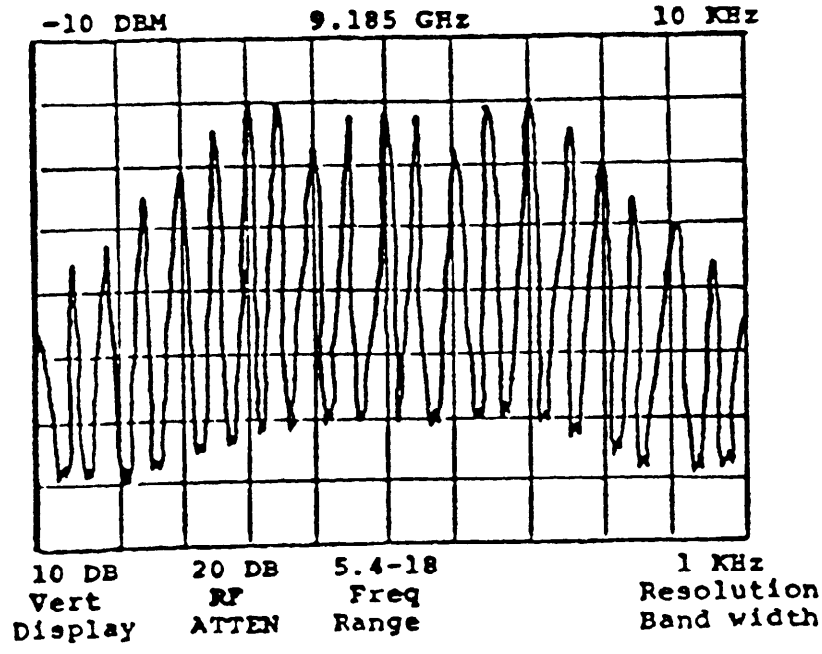


FIGURE 9. Frequency modulated signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

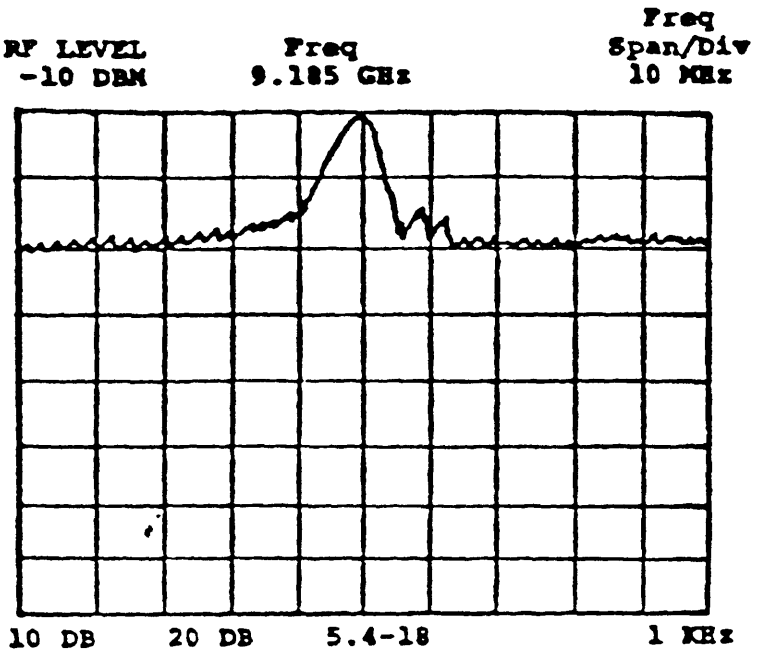


FIGURE 10a. Pulse modulated signal.

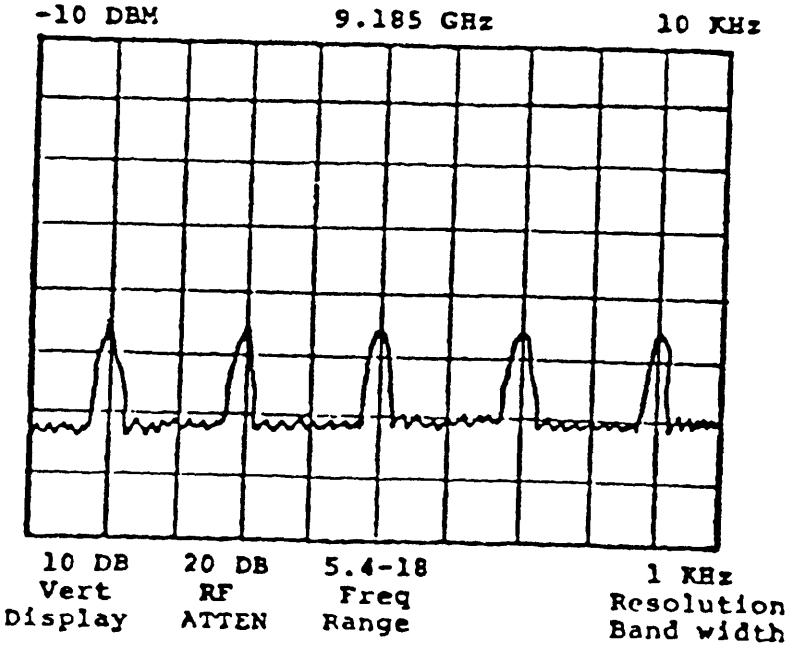


FIGURE 10b. Pulse modulated signal.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:36Z
Check the source to verify that this is the current version before use.

4.7.7.12 Power output and microwave chassis loss.

a. Connect power meter (item 12, Table V) with 20 db attenuator (item 11, Table V) to J3 (TRANSMIT PULSE OUT). With conditions in 4.7.3.llb and attenuator AT-1 set at O db, power meter shall read 0.30 mw minimum indicating a minimum power output of 30 rnw on all channels.

b. With conditions in 4.7.3.12a and CHANNEL SELECTOR switch set to channel 7, measure and record reference power output on J3 (TRANSMIT PULSE OUT).

Connect power meter (Item 12, Table V) to SPRCTRUM ANALYZE jack (J24). Power output will be 27 db to 30.5 db below the reference value, with a termination (item 32, Table V) connected to J3.

d. Connect power meter (item 12, Table V) to STALO MONITOR (J9). Power reading will be between 15.0 db to 20.5 db below the reference value, with a termination (item 32, Table V) connected to J3.

Connect power meter (item 12, Table V) to TRANS PULSE OUT jack (J3). With MOD MODE switch to OFF, power output shall be no greater than 40 db below the reference value.

f. Using a frequency source (item 17, Table V) of between 9.145 GHz and 9.245 GHz, insert a 50 mW $\pm$ 1% signal into J1 ATTENUATOR jack. Connect power meter and 20 db attenuator (Item 11, Table V) to J3 of tester panel. MOD MODE switch to ON. Depress S6 to illuminate EXTERNAL legend. Minimum power reading shall be .19 mW with AT 1 set of O db.

g. Rotate AT-1 control slowly between O db and maximum settings. Power reading shall decrease continuously. The maximum variation between AT-1 reading and measured attenuation shall be $\pm$ 1 db.

h. Measure dc output using item 1 of Table V at jack J3. Check dc level for conformance to 3.5.9.2 requirements.

4.7.3.13 Calibration of MX-8046A/TPM-22A noise source.

a. Connect apparatus as shown in Figure 11. Before operation, set waveguide attenuator (item 30, Table V) so that the output of the X-band source is 10 mW measure with power meter (Item 12, Table V) and attenuator (item 4, Table V). Supply power to all units and set noise source current such that the noise figure is 25 db. Record power meter reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

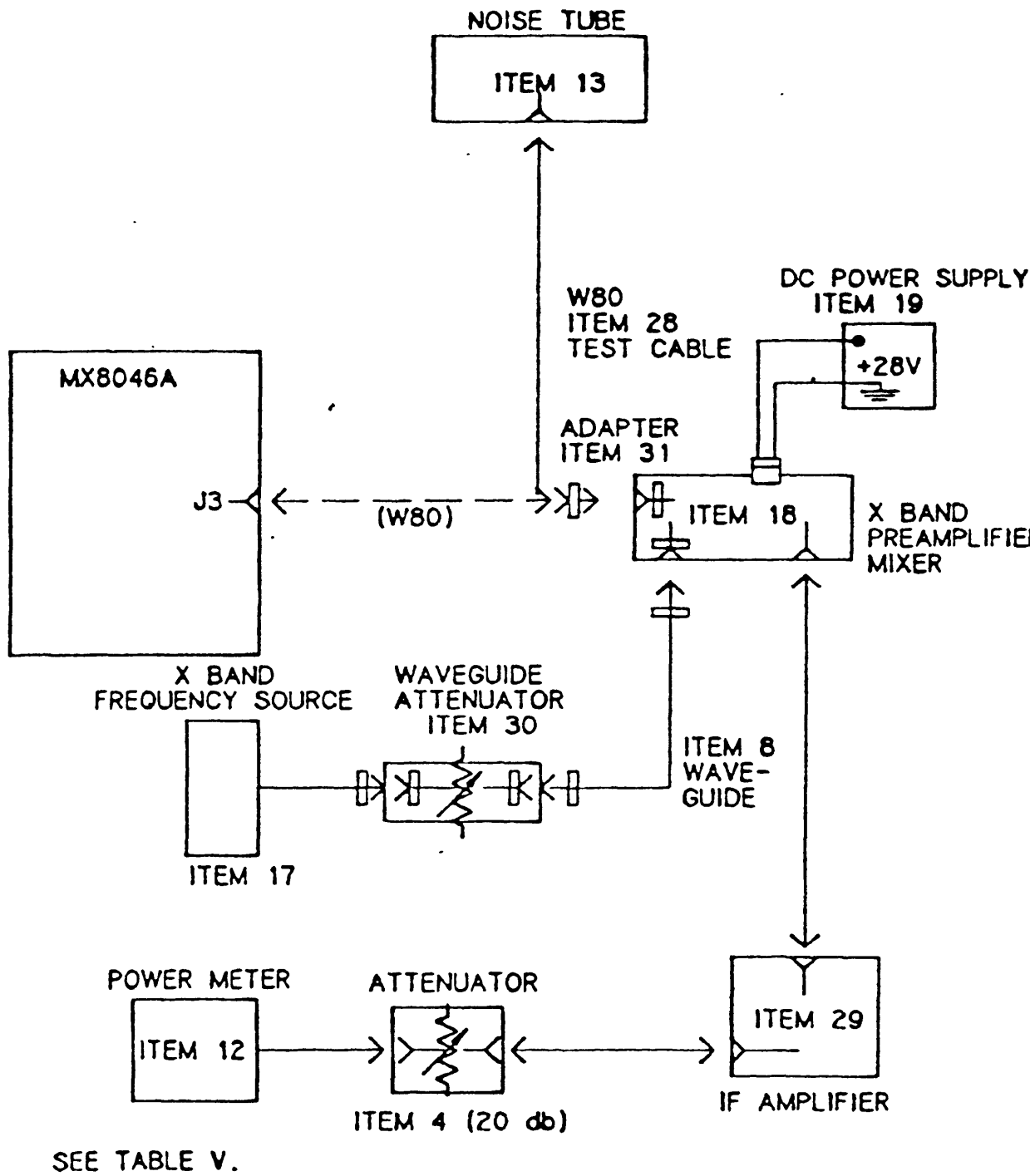


FIGURE 11. Calibration test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

b. Remove test cable (item 28, Table V) from noise source and connect J3 of MX-8046A. Depress S3 of MX-8046A to Illuminate NOISE FIGURE legend and note power meter reading.

If power meter differs from reading recorded in 4.7.3.13a by more than ± 0.5 db, locate miniature attenuator A2AT-1 on MX-8046A microwave chassis and adjust until power meter reading agrees with that recorded in 4.7.3.13a.

4.7.3.14 MOD TRIGGER. 60 MHz, and TO connections. MOD TRIGGER Jll, J15, J5-A1. Attach BNC Tees (item 5, Table V) to Jll, and J15 to the TEE at Jll connect signal generator (Item 10, Table V) and channel 1 of oscilloscope (item 2, Table V). To TEE at J15 connect 95 ohm load (item 14, Table V). Set signal generator to produce a 20 KHz square wave signal. Using a probe (item 16, Table V) connected to CH2 of oscilloscope, compare the signal at J15 and J5-A1 to that on Channel 1 to ascertain that there is no amplitude or phase distortion.

4.7.3.15 60 MHz CW J12 and J16. Connect BNC TEEs (item 5, Table V) to J12 and J16. To TEE and J12 connect 60 MHz sourct 3, Table V) and Channel 1 of oscilloscope. To the TEE at J16 connect 50 ohm load (item 15, Table V) and Channel 2 of oscilloscope. Set sweep on oscilloscope so that actual 60 KHz waveform is discernible to ascertain that there is no amplitude or phase distortion between the signal at J12 and that at J16.

4.7.3.16 TO TRIGGER J14, J17, J18. Perform same test as described in 4.7.3.14 with input at J14 and 50 ohm loads (item 15, Table V) at both J17 and J18.

4.7.3.17 +24 Vdc control out. With conditions as described in 4.7.3.1 measured with voltmeter (item 1, Table V) there shall be +24 ± 4 Vdc across item 27 (see Figure 4).

4.8 soldering. Soldering processes shall be inspected by certified inspectors to the requirements specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 11739273. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The receiver/transmitter tester is used for testing, aligning and troubleshooting the Receiver/Transmitter, Unit 2 of Radar Set AN/VPS-2 and some of its repairable assemblies and subassemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Applicable stock number.

c. First article sample requirements.

d. Packaging requirements, if other than specified In Section 5.

e. Serialization requirements, if applicable.

f. Certification of conformance for each lot of shipment of product.

g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, Attn: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Radar Set AN/VPS-2
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-Al?

(Project 4931-A289)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisi letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
3. DOCUMENT TITLE		

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:36Z
Check the source to verify that this is the current version before use.